

EBN111 Klastoets/Class Test#1: 1 Maart/March 2013

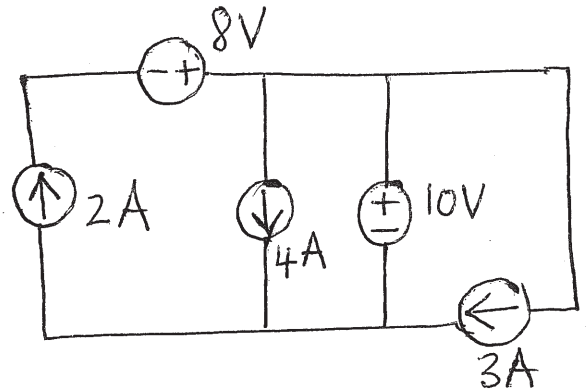
- Q1 'n Nuwe soort battery lewer 10W drywing vir 8 ure sonder enige variasie in spanning of stroom. Na 8 ure val die drywingsuitset lineêr van 10W na 0 binne 30 minute. Bepaal die energie stoorkapasiteit van die battery. [2]
A new type of battery delivers 10W of power for 8 hours without voltage or current fluctuation. After 8 hours the power output drops linearly from 10W to 0 in 30 minutes. Determine the energy storage capacity of the battery.

$$w=297\text{kJ}$$

- Q2 Bereken die drywing ge-assosieer met die 8V spanningsbron en die 3A stroombron. [2]
Calculate the power associated with the 8V voltage source and the 3A current source.

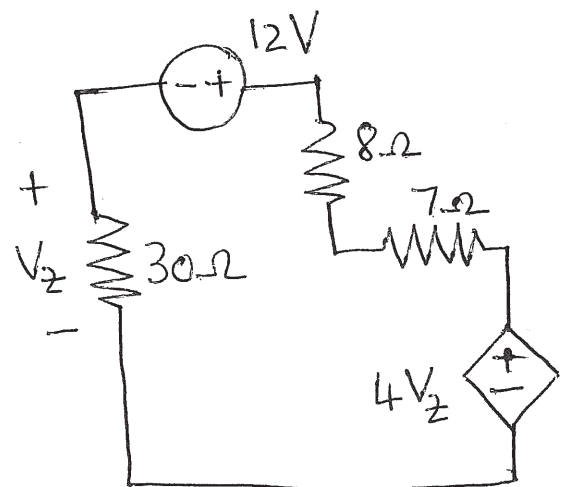
$$P_{8V}=-16\text{W}$$

$$P_{3A}=30\text{W}$$



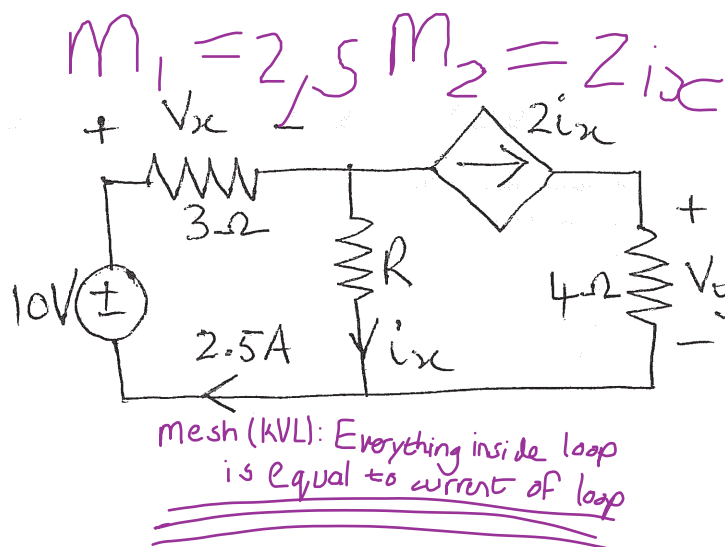
- Q3 Bepaal v_z in die stroombaan. [2]
Determine v_z in the circuit.

$$V_z=4.8\text{V}$$



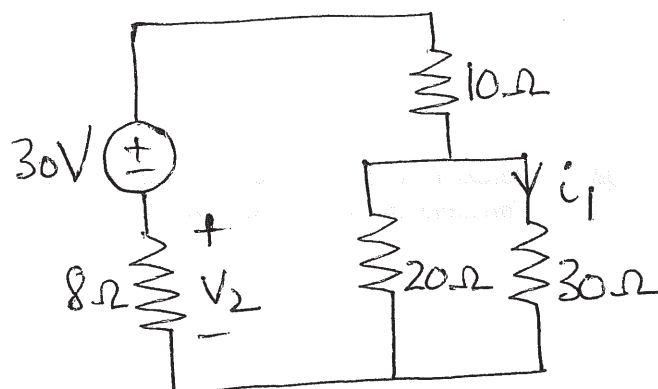
- Q4 Bereken v_x , v_y en i_x in die stroombaan. [3]
Calculate v_x , v_y and i_x in the circuit.

$v_x = 7.5V$
 $v_y = 6.664V$
 $i_x = 0.833A$



- Q5 Bereken/Calculate
 (a) i_1 [2]
 (b) v_2 [2]

$i_1 = 0.4A$
 $v_2 = -8V$



- Q6 Bereken R_{eq} . [2]
 Calculate R_{eq} .

$R_{eq} = 20 \text{ ohms}$

